Architecture and Core Concepts

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Storefront Next Architecture

Storefront Next is a full-stack React framework for building B2C Commerce storefronts. It combines server-side rendering with client-side navigation to deliver performant and personalized shopping experiences. Storefront Next runs on Managed Runtime, the infrastructure for deploying, hosting, and monitoring storefronts.

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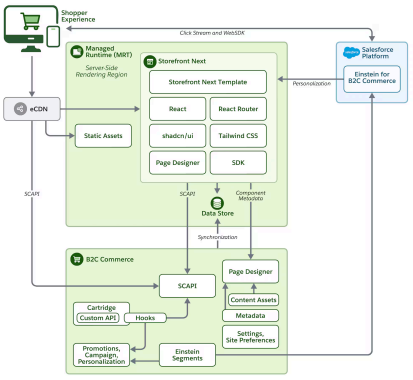
Architectural Components of Storefront Next

This diagram shows the architecture of Storefront Next, including the request flow from the Shopper Experience to Managed Runtime, data retrieval from the B2C Commerce instance, and interaction with the Salesforce Platform.



Here's a description of the request flow.

1. Frontend request: The shopper sends a request to Managed Runtime, which handles server-side rendering (SSR) via the Storefront Next app.
2. Data Store fetching: The Storefront Next app calls into the Managed Runtime Data Store to fetch any required Site Preferences for rendering.

4. Business logic: The B2C Commerce system executes cartridge logic. It accesses data, such as products, pricing, and content, and uses configurations from Business Manager.
5. Personalization and recommendations: The storefront communicates with the Salesforce Platform via Recommendations APIs to get product recommendations. It fetches personalized content powered by the personalization engine.

Managed Runtime and Storefront Next

Managed Runtime provides the infrastructure to deploy, host, and monitor the Storefront Next app. Managed Runtime includes these services.

- Static asset storage: Enables uploading and deploying your static assets, such as code artifacts and images.
- Data storage: Enables synchronizing and storage of site preferences from Business Manager for rendering.
- App server: A scalable serverless Node.js runtime environment for your storefront. The Storefront Next app runs on the Managed Runtime app server and uses these technologies: React 19, React Router 7, shadcn/ui, and Tailwind CSS.

B2C Commerce Backend

The B2C Commerce platform provides the ecommerce services and features that your storefront consumes.

- Cartridge: Server-side extensions including custom APIs and hooks.
- Business Manager: Where merchants manage content assets, promotions, campaigns, and personalization.
- Einstein Segments: Customer segmentation for personalized experiences.
- Page Designer: Page Designer connects to the storefront via component metadata. Merchants use Page Designer, a visual editor in Business Manager, to design, schedule, and publish custom pages in the storefront

Integration

The storefront connects to the Salesforce Platform through these services.

- Personalization Engine: Delivers personalized content and recommendations.
- Einstein for B2C Commerce: AI-powered product recommendations. Data flows between B2C Commerce and the Salesforce Platform through segment synchronization.

Core Foundation: React Router 7

Storefront Next is built on React Router 7 in framework mode. Before diving into Storefront Next specifics, understand this foundation.

React Router 7 is more than a client-side router–it's a full-stack framework that handles:

- Server-side rendering: Pages render on the server and stream HTML to the browser.
- Data loading: Loaders fetch data before components render.
- Mutations: Actions handle form submissions and data changes.
- Client navigation: After initial load, navigation happens without full page reloads.

If you're familiar with Remix, React Router 7 framework mode uses the same patterns. If you're new to these concepts, we recommend reviewing the React Router Documentation.

Technology Stack

Technology	Purpose	Highlights
React 19	UI components and rendering	Includes Node.js streaming, enabling data to be read in small chunks and processed incrementally which offers

React Router 7	Full-stack framework (routing, SSR, data loading)	React Router's Framework Mode provide a comprehensiv development experience by integrating server-side rendering (SSR) and dat handling capabilities directly withir React Router.
Vite	Build tool and development server	Provides fast server start times and improved Hot Module Replacement (HMR) performance.
TypeScript	Statically Type Programming Language	Enables defining types at design time and catching type-related errors early during development.
Tailwind CSS	Styling	Optimized for performance by eliminating runtime overhead, supporting features like tree-shaking and partial rendering in modern React apps.
shadcn/ui	Open-source library for accessible UI components	Ensures high accessibility standards (WCAG/ARIA) and full customizatior through the underlying Tailwind CSS.

Page Rendering

Storefront Next uses React Router's rendering model to deliver fast, personalized pages. Understanding this model helps you

1. Initial page load: The first page load uses streaming SSR for data fetching, which ensures that personalization data, such as pricing and promotions, is included in the HTML payload. With React's streaming model, static content appears immediately while dynamic fragments fill in as data becomes available.
2. Subsequent navigations: After the initial load, Storefront Next fetches data via the server-side data loaders. Navigations are handled client-side using React Router 7, which minimizes network requests and enables smooth transitions.

Initial Page Load

When a shopper first visits your storefront:

1. The request reaches Managed Runtime after passing through eCDN.
2. Middleware runs to set up authentication, configuration, and internationalization.
3. The route's loader fetches data from the Salesforce Commerce API.
4. React renders components on the server and streams HTML to the browser.
5. The browser shows content as it arrives.
6. React hydrates the page, making it interactive.

```
Browser Request
      ↓
   eCDN (cache check)
      ↓
Managed Runtime (Lambda)
      ↓
   Middleware Chain
   ├── App Config
   ├── i18n Setup
   ├── Authentication
   └── Analytics
      ↓
   Route Loader
   (fetches data from Commerce API)
      ↓
   React Render
   (streams HTML)
      ↓
```

Navigation

After the browser receives the streamed HTML, React hydrates the page. Hydration attaches event handlers to the server-rendered HTML, making buttons clickable, forms submittable, and components interactive. The page is visible immediately and becomes fully interactive after hydration completes.

After hydration, subsequent navigation happens entirely on the client.

1. React Router intercepts link clicks.
2. The new route's `loader` fetches needed data.
3. React updates the DOM without a full page reload.

This hybrid approach gives you the SEO and performance benefits of server rendering with the smooth experience of a single-page app.

Next Steps

To dive deeper into the technical aspects of Storefront Next, check out these links.

- Project Configuration and Build Tools
- Data Fetching
- Loading States
- State Management
- Basket Management
- SCAPI Client
- Content Caching
- Storage and Sessions
- Shopper Context
- Server API Routes
- UI Styling
- Images
- Static Assets
- SEO and Metadata
- AEO and GEO
- Multisite Configuration
- Internationalization
- Adapters
- Security Response Headers